Do Now · retrieval

Noughts & Crosses · Part 4 — Complete and test

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does MakeMove() return after a successful move?

Q2

For a 3 × 3 C# array, what does board.GetLength(0) return?

Q3

In Triple(7) , identify the argument. In Triple(int number) , identify the parameter.

Q4

A 3 × 3 game board is displayed using repeated row-printing statements. Explain how nested loops improve the code and what each loop controls.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

HasWon() checks 8 lines — CODE:
3 rows
3 columns
2 diagonals
_____

8 winning lines — Each winning line needs three comparisons joined with && . The possible lines are joined with || .

Core idea 2

BoardFull() looks for any free square — CODE:

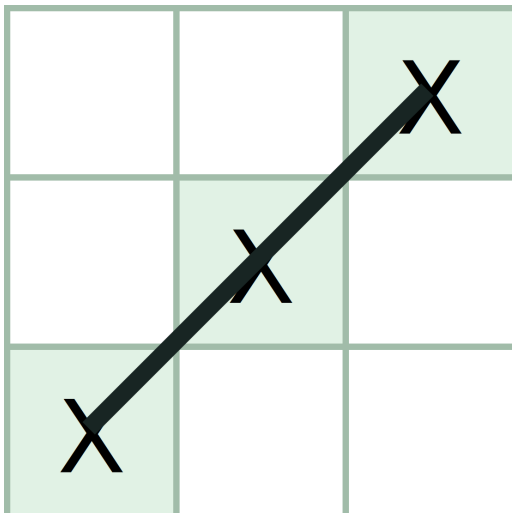
```
if (board[row, col] != "X" &&
board[row, col] != "O")
{
return false;
}
```

return true; — If even one square is free, the board is not full.

Core idea 3

Finish Main() — CODE:
valid move
→ check win
→ check draw
→ otherwise switch player — Part 3 already handled valid moves and switching. Today we add the stopping conditions.

Key visual · complete concept



New Learning · applying the ideas

Core idea 4

Can two pupils actually play it? — After completing the code, play against a partner and compare expected with actual results. Use planned board positions to test all eight winning lines.

Vocabulary · use the correct technical terms

Boolean — A data type with only two possible values: true or false.

Condition — An expression that evaluates to true or false.

State — The current data describing what is happening in the program.

Game loop — The repeated structure that controls turns until the game ends.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Build BoardFull()

Follow the worked method

1. Use nested loops — CODE:

```
for (int row = 0; row < board.GetLength(0); row++)
{
for (int column = 0; column < board.GetLength(1); column++)
{
}
}
```

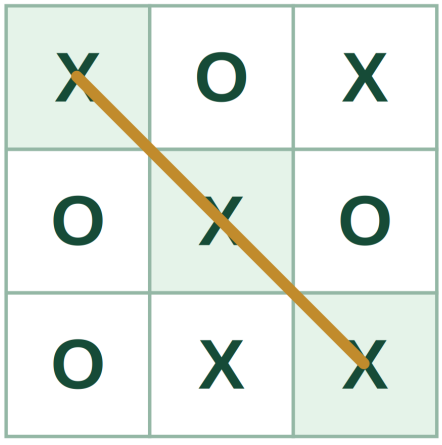
2. No free square was found — CODE:

return true; — This line is reached only after every position has been checked.

3. Check draw after checking win — CODE:

```
if (HasWon(currentPlayer))
{
// win
}
else if (BoardFull())
{
// draw
}
```

Source model · final illustrated step



Full board AND X has won

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Complete the condition
CODE:

```
if (board[0,0] == ? &&
board[1,1] == ? &&
board[2,2] == ?)
{
return true;
}
```

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Finish all 8 winning lines
Continue your Lesson 015 program.
Complete HasWon(string player) so it detects all 3 rows, all 3 columns and both diagonals.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Partner challenge: play and test
Pair up: one pupil plays X and the other O. Run your own C# program, make moves 1–9 and swap who goes first for a second complete game.
Evidence: for each targeted test, record the input or board state, expected result, actual result and pass/fail. Fix and retest any failure. Test all eight winning lines using prepared moves or board states; ordinary matches alone may not cover them.
If your online C# runner does not support repeated keyboard input, run the program in a local C# console or use planned input sequences for testing.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

After a valid move, the program should check HasWon() before BoardFull() .

Use key vocabulary from memory

1. Explain Boolean in your own words.

2. Explain Condition in your own words.

3. Explain State in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.